Multi-probe cosmology in September 2026: a reproducible MCMC analysis of Planck, ACT, DESI and DES-Dovekie

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ABSTRACT

We present newly executed cosmological parameter inference using public CMB, baryon acoustic oscillation (BAO) and Type Ia supernova likelihoods. Our baseline combines Planck 2018 large-scale temperature and polarization, ACT DR6 foreground-marginalized spectra, DESI DR2 galaxy BAO, the July 2026 Lyα Alcock–Paczynski plus BAO distance measurement, and the recalibrated DES-Dovekie supernova sample. We fit spatially flat ΛCDM, constant-w CDM and Chevallier–Polarski–Linder (CPL) dark energy, varying the six standard CMB parameters and instrumental calibration parameters. In ΛCDM, we obtain $H_0 = 68.26 \pm 0.27\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.3027 \pm 0.0036$. CPL gives $w_0 = -0.83 \pm 0.06$ and $w_a = -0.61 \pm 0.23$, with a local deviance improvement of 10.44 for two extra parameters. Four independent Markov chains per model sample a numerical theory surrogate; retained draws are recalculated with CAMB and importance-corrected to the direct likelihood. We report both chain convergence and correction efficiency. Pantheon+ and the previous Lyα BAO measurement provide sensitivity tests. The results are conditional on fixed neutrino mass and standard early-time physics. Recent weak-lensing, cluster, SPT and distance-scale measurements are discussed as external comparisons, rather than incorporated through unvalidated products of published parameter estimates.

Key words: cosmological parameters – dark energy – cosmic background radiation – large-scale structure of Universe – supernovae: general – methods: statistical

1 INTRODUCTION

The present cosmological data set probes the same expansion history in markedly different ways. CMB acoustic structure calibrates the physical matter densities and the sound horizon. BAO traces radial and transverse distances relative to that ruler. Standardized supernovae measure the relative luminosity-distance relation. Agreement among these observables is a stronger test than agreement among parameter values obtained after imposing ΛCDM separately.

Two recent changes motivate a fresh calculation. DES-Dovekie updates the cross-survey photometric calibration and light-curve analysis of DES-SN5YR (Popovic et al. 2026). DESI's July 2026 Lyα analysis adds information from the Alcock–Paczynski effect to its BAO distances (DESI Collaboration et al. 2026). Both are relevant to claims of evolving dark energy. Meanwhile, ACT DR6 provides precise CMB polarization data that complement Planck (Louis et al. 2025; Calabrese et al. 2025).

This paper answers a deliberately specified statistical question: what do these public likelihood products imply within three nested flat cosmological models? It is a reanalysis of released summary likelihoods, not a new reduction of detector maps, galaxy catalogues or supernova light curves. Table 1 states the fitted scope. Other major contemporary experiments are assessed in Section 5; they are not silently treated as independent Gaussian parameter priors. All tables identified as our results derive from calculations executed for this draft.

2 COSMOLOGICAL MODELS AND LIKELIHOODS

2.1 Theory and priors

We assume spatial flatness, adiabatic scalar perturbations, a power-law primordial spectrum, standard recombination and general relativity. CAMB (Lewis et al. 2000) calculates the expansion, lensed spectra, sound horizons and linear matter fluctuation amplitude. The varied CMB parameters are $\omega_b = \Omega_b h^2$, $\omega_c = \Omega_c h^2$, $100\theta_\star$, $\ln(10^{10}A_s)$, n_s and τ. We derive H_0, Ω_m, r_d, the cosmic age, σ_8 and $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$. The sum of neutrino masses is fixed to 0.06 eV, with one massive species and $N_\mathrm{eff} = 3.046$. Helium follows CAMB's BBN consistency relation; no primordial-abundance likelihood is added. Tensors, running and spatial curvature are fixed to zero, and lensing is not assigned a free phenomenological amplitude. The primordial pivot is $k = 0.05\,\mathrm{Mpc^{-1}}$. Non-linear corrections to the lensing calculation use CAMB's HMcode-2020 prescription with its default parameters; no separate baryonic-feedback parameter is fitted.

For ΛCDM, $w = -1$. Constant-w CDM adds one parameter. CPL adds two,

$$w(a) = w_0 + w_a(1-a),$$
$$\frac{\rho_\mathrm{de}(a)}{\rho_\mathrm{de,0}} = a^{-3(1+w_0+w_a)}\exp[-3w_a(1-a)]. \quad (1)$$

following Chevallier & Polarski (2001); Linder (2003). The parametrized post-Friedmann treatment permits crossing $w = -1$. Sampling uses $w_p = w_0 + w_a/4$ to reduce correlations; the linear transformation has unit Jacobian and does not change a flat (w_0, w_a)

Table 1. Data scope and treatment. CMB bandpowers retain their released covariance and windows. Counts refer to retained likelihood entries, not numbers of independent sky modes.

Probe	Fitted information	Treatment and limitation
Planck 2018	252 Plik-lite bins: TT to $\ell = 1000$, TE/EE to 600; Commander TT and SimAll EE at $2 \le \ell \le 29$	Foregrounds marginalized upstream; shared CMB calibration with Gaussian 0.25 per cent prior.
ACT DR6	135 TT/TE/EE bandpowers, $600 \le \ell_{\rm eff} \le 6500$	Supplied windows through $\ell = 8501$; polarization efficiency varied.
DESI DR2	13 BAO distances from $z = 0.295$ to 2.33	Full released covariance; no additional SDSS/DESI DR1 duplication.
DESI Lyα 2026	Joint $D_M/r_{\rm d}, D_H/r_{\rm d}$ at $z = 2.33$	Replaces older Lyα pair; rounded Gaussian summary from equation (26) of DESI Collaboration et al. (2026).
DES-Dovekie	1820 supernovae, total released precision	Individual redshifts; common magnitude offset marginalized.
Pantheon+	1590 light curves at $z_{\rm HD} > 0.01$	Alternative to Dovekie; full covariance; no Cepheid calibration.
Other recent probes	DES Y6, KiDS-Legacy, SPT-3G, lensing reconstruction, clusters, local distances, sirens	Literature context only; no raw likelihood refit in this work.

prior. This fixed w_p is the value at $z = 1/3$, not a data-optimized pivot.

The uniform prior ranges are $\omega_b \in (0.0205, 0.0245)$, $\omega_c \in (0.105, 0.135)$, $100\theta_\star \in (1.038, 1.044)$, $\ln(10^{10}A_s) \in (2.8, 3.3)$, $n_s \in (0.92, 1.03)$, $\tau \in (0.01, 0.12)$, $w_0 \in (-2, 0)$ and $w_a \in (-3, 1)$, with $w_0 + w_a < 0$ and CAMB solutions restricted to $35 < H_0/{\rm km\,s^{-1}\,Mpc^{-1}} < 100$. We inspect posterior occupancy near these limits. Instrument parameters are $A_{\rm CMB} \sim \mathcal{N}(1, 0.0025^2)$, restricted to $(0.98, 1.02)$, and $P_{\rm ACT} \in (0.9, 1.1)$.

2.2 CMB

The Planck likelihood combines Plik-lite with the native Commander and SimAll implementations distributed through Cobaya (Planck Collaboration et al. 2020). We retain Plik-lite bins whose upper edges lie below 1000 in TT and 600 in TE and EE, following the large-scale Planck plus ACT construction. Covariance submatrices are selected before inversion. ACT DR6 uses the collaboration's foreground-marginalized TT/TE/EE data (Louis et al. 2025). Its Gaussian term is

$$-2\ln\mathscr{L}_{\rm ACT} = (d - WD_\ell)^T C_{\rm ACT}^{-1}(d - WD_\ell) + {\rm const.}, \tag{2}$$

where $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$. Theory is divided by $A^2_{\rm CMB}$ and, for ACT TE and EE, additionally by $P_{\rm ACT}$ and $P^2_{\rm ACT}$. Low-multipole likelihoods retain their non-Gaussian form.

There remains TT multipole overlap between Planck and ACT at $600 \lesssim \ell \lesssim 1000$; we follow the public combination's approximation of neglecting cross-experiment covariance. Foreground marginalization, these cuts, SimAll rather than SRoll2, and the omission of reconstructed CMB-lensing likelihoods distinguish our results from the full published P-ACT combinations. We have not reproduced the multi-frequency foreground analysis or independently validated its compression for every dark-energy extension.

2.3 BAO and the 2026 Lyα update

We use the DESI DR2 mean vector and covariance (Abdul-Karim et al. 2025), comparing to

$$D_H = \frac{c}{H(z)}, \quad D_M = (1+z)D_A, \quad D_V = [zD_M^2 D_H]^{1/3}, \tag{3}$$

in units of the CAMB drag-epoch sound horizon. To include the new full-shape Lyα information, we replace the old pair at $z = 2.33$ with

$$D_M/r_{\rm d} = 39.32 \pm 0.33, \tag{4}$$

$$D_H/r_{\rm d} = 8.600 \pm 0.066, \qquad \rho = 0.225. \tag{5}$$

The covariance is reconstructed from these published rounded errors and correlation (DESI Collaboration et al. 2026). This uses the released distance-level result; it does not refit Lyα correlation functions. Appending this pair to the original BAO pair would double-count the same forest data. The original BAO-only pair is retained for a replacement sensitivity test.

2.4 Supernovae

DES-Dovekie supplies a total *precision* matrix in packed upper-triangular form, rather than a covariance requiring a further diagonal statistical term (Popovic et al. 2026). We reconstruct it without changing the release ordering. Pantheon+ uses the total covariance after applying $z_{\rm HD} > 0.01$ (Brout et al. 2022; Scolnic et al. 2022); repeated light curves are retained with their correlations. These compilations are fitted as alternatives because they share objects and calibration information.

Using both the Hubble-diagram and heliocentric redshifts, the model modulus is

$$\mu_i = 5\log_{10}\left[(1+z_{{\rm HD},i})(1+z_{{\rm hel},i})D_A(z_{{\rm HD},i})/{\rm Mpc}\right] + 25. \tag{6}$$

For residual vector Δ, precision P and an uninformative common magnitude-offset prior, the parameter-dependent marginalized statistic is

$$\chi^2_{\rm SN} = \Delta^T P \Delta - \frac{(1^T P \Delta)^2}{1^T P 1}. \tag{7}$$

The fixed offset-normalization term cancels between models using the same sample. No SH0ES absolute calibration or independent Gaussian H_0 constraint enters these fits. Supernovae alone therefore do not measure H_0 here.

3 COMPUTATION AND VALIDATION

Direct theory evaluation through the ACT window support is expensive. We fit a local cubic polynomial to a bank of 640 CAMB predictions in eight cosmological coordinates. The approximation

includes spectra and distance predictions. A 100-node log-redshift interpolation accelerates the supernova term while retaining its full quadratic form. This is numerical interpolation of a smooth prediction, not statistical replacement of the supernova data by 100 independent objects.

Each model is sampled with four separately seeded Metropolis chains. Proposal covariances and scales adapt during 5000 discarded steps and are frozen for 30,000 production steps per chain. We inspect trace behaviour, rank-normalized folded/split $\widehat{R}$, bulk and tail effective sample sizes (Vehtari et al. 2021). Convergence of the proposal posterior alone does not establish accuracy of cosmological constraints.

We therefore select draws across all four chains, calculate their spectra and individual-supernova distances with CAMB, and apply

$$u_i = \exp[\ln \mathscr{L}_{\rm direct}(\theta_i) - \ln \mathscr{L}_{\rm surrogate}(\theta_i)], \quad \tilde{u}_i = \frac{u_i}{\sum_j u_j}. \tag{8}$$

This importance correction follows the general approach described by Lewis & Bridle (2002). The final parameter tables and credible regions use these direct-likelihood weights. We record $N_{\rm IS} = 1/\sum_i \tilde{u}_i^2$ and Pareto-tail diagnostics separately from MCMC ESS. Raw surrogate chains, retained direct evaluations and both log densities are included in the reproducibility bundle. All models satisfy $\widehat{R} \leq 1.0030$ and minimum bulk ESS 3071. We use 800 stratified direct evaluations per model. Forty contiguous batches of the importance-sampling influence function give a maximum mean Monte Carlo error of 4.5 per cent of a posterior standard deviation. These estimates account for residual correlation between nearby selected draws. The dark-energy posterior lies well inside the stated prior boundaries.

CAMB version 2.0.4 is used with Recfast recombination, lensed spectra through $\ell = 8501$, lens-potential accuracy 1, accuracy boosts 1.2 and a lens-output margin of 700. A higher-accuracy calculation uses boosts 2, lens-potential accuracy 2 and margin 1200. At the three fitted modes, the squared high-ℓ CMB plus BAO data-space difference between these settings is 0.028–0.030. The common absolute deviance changes by approximately one, while the largest change in a between-model deviance difference is 0.078. These fixed-point checks are not a new high-accuracy MCMC run. An independent low-redshift quadrature reproduces the SN-only Dovekie constraint, $\Omega_m = 0.3308 \pm 0.0153$, and gives 0.3323 ± 0.0182 for Pantheon+. These checks address numerical error; they do not remove physical modelling or inter-survey calibration uncertainty. The Gaussian high-ℓ terms and SN residuals are inspected separately. Because low-ℓ terms are non-Gaussian and contain likelihood offsets, their sum is not assigned a naive global χ^2 goodness-of-fit probability.

4 RESULTS

4.1 Parameter constraints

Table 3 and Fig. 1 show the corrected baseline constraints. The ΛCDM expansion rate is $H_0 = 68.26 \pm 0.27\,{\rm km\,s^{-1}\,Mpc^{-1}}$, with $S_8 = 0.812 \pm 0.009$. Constant-w gives $w = -0.976 \pm 0.021$, close to -1. CPL favours $w_0 = -0.830 \pm 0.058$ and $w_a = -0.609 \pm 0.228$. The two dark-energy coordinates are strongly correlated. Figure 3 expresses the resulting change as a redshift-dependent equation of state and distance residuals. Allowing CPL lowers the inferred expansion rate to $H_0 = 67.45 \pm 0.57\,{\rm km\,s^{-1}\,Mpc^{-1}}$ in this combination.

4.2 Fit quality and model comparison

The best evaluated direct-CAMB local deviances improve by 1.06 and 10.44 relative to ΛCDM for constant-w and CPL, respectively. Each mode begins at the surrogate optimum. A finite-difference direct-likelihood correction is used in a local Powell search, and the candidate is reevaluated with CAMB; this does not certify a global maximum. The resulting fit-statistic decomposition is in Table 4; the CMB residuals appear in Fig. 2. The CPL improvement is distributed across the supernova, BAO and ACT terms, rather than coming from a single added likelihood. We report an AIC-like penalized-deviance diagnostic, $\Delta D + 2\Delta k$, for the nested models with unchanged data and calibration treatment. It is not a Bayesian evidence calculation. The common SN-offset normalization cancels, and the additional cosmological parameter counts are one and two for constant-w and CPL. We do not convert $\sqrt{\Delta D}$ into a Gaussian significance for a two-parameter extension. Neither a posterior credible contour nor this diagnostic establishes a discovery of a physical dark-energy field.

4.3 Data-choice sensitivity

Table 5 gives the adequately sampled replacement tests. For CPL, reverting to the original Lyα pair shifts the mean w_0 by $+0.015$ and w_a by -0.060. For CPL, replacing Dovekie by Pantheon+ shifts the mean w_0 by -0.034 and w_a by $+0.113$. Supernovae omitted in wCDM has $N_{\rm IS} = 35$ (Pareto $k = 0.92$), so we do not report parameter constraints for that reweighting. Supernovae omitted in CPL has $N_{\rm IS} = 129$ (Pareto $k = 0.61$), so we do not report parameter constraints for that reweighting. The accompanying diagnostics provide paired Monte Carlo errors for every parameter shift, using the same direct-evaluation draws in each comparison. These replacement tests preserve the CMB treatment and cosmological priors. They assess sensitivity, not the independence of overlapping surveys. When importance weights become concentrated, the resulting effective sample count limits the precision of the comparison; a small weight ESS is not repaired by quoting the much larger original chain length.

5 RELATION TO THE WIDER DATA LANDSCAPE

The following are published measurements, not new joint fits performed here. DES Y6 reports $S_8 = 0.789 \pm 0.012$ in flat ΛCDM and finds a 1.8σ full-parameter difference from its Planck–ACT–SPT reference, versus 2.6σ when projected onto S_8 (DES Collaboration et al. 2026). KiDS-Legacy gives $S_8 = 0.815^{+0.016}_{-0.021}$, compatible with its Planck comparison (Wright et al. 2025). The differing calibrations and nuisance treatments caution against describing a unique, experiment-independent "S_8 tension". Our inferred S_8 provides a conditional comparison, not a new weak-lensing measurement or a growth-versus-geometry consistency fit.

SPT-3G D1 is a major further CMB test (Camphuis et al. 2025); adding it requires its likelihood, calibration and overlap prescription. Reconstructed ACT/Planck CMB lensing (Madhavacheril et al. 2024) tests growth and helps break geometric degeneracies, but shares sky information with primary CMB data. Cluster abundance, including eROSITA (Ghirardini et al. 2024), adds mass-calibration, selection and non-linear modelling requirements. We do not replace these analyses with one-dimensional S_8 Gaussian likelihoods in extended cosmologies.

The Local Distance Network reports $H_0 = 73.50 \pm 0.81\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (H0DN Collaboration et al. 2026). Our inverse-distance-ladder constraints can be compared with that

Table 2. Production-chain and importance-correction diagnostics. Four chains per model each contain 30,000 production steps after 5000 warmup steps. Chain diagnostics cover physical and derived coordinates. N_{mIS} describes the direct-evaluation weights and is distinct from MCMC ESS.

Model	Max. $\widehat{R}$	Min. bulk ESS	Min. tail ESS	Direct draws	$N_{\rm IS}$	Pareto k
ΛCDM	1.0014	3925	7140	800	799	-0.01
wCDM	1.0030	3636	6228	800	797	0.00
CPL	1.0016	3071	4950	800	789	0.47

Table 3. Direct-CAMB importance-corrected posterior means and standard deviations for the baseline combination. Equal-tailed 16th, 50th and 84th percentiles and Monte Carlo errors are supplied in the accompanying tables. Fixed quantities have no quoted uncertainty.

Parameter	ΛCDM	wCDM	CPL
ω_b	0.02255 ± 0.00010	0.02257 ± 0.00011	0.02249 ± 0.00011
ω_c	0.1179 ± 0.0007	0.1173 ± 0.0008	0.1190 ± 0.0009
$100\theta_\star$	1.04110 ± 0.00023	1.04114 ± 0.00025	1.04098 ± 0.00026
$\ln(10^{10}A_s)$	3.050 ± 0.015	3.051 ± 0.016	3.047 ± 0.016
n_s	0.9740 ± 0.0030	0.9751 ± 0.0034	0.9715 ± 0.0033
τ	0.0594 ± 0.0074	0.0604 ± 0.0079	0.0564 ± 0.0077
H_0 [km s^{-1} Mpc^{-1}]	68.26 ± 0.27	67.73 ± 0.51	67.45 ± 0.57
Ω_m	0.3027 ± 0.0036	0.3064 ± 0.0046	0.3126 ± 0.0055
σ_8	0.8082 ± 0.0065	0.8000 ± 0.0099	0.8106 ± 0.0101
S_8	0.8119 ± 0.0090	0.8085 ± 0.0095	0.8274 ± 0.0110
r_d [Mpc]	147.46 ± 0.20	147.58 ± 0.22	147.22 ± 0.24
Age [Gyr]	13.770 ± 0.014	13.780 ± 0.016	13.742 ± 0.020
w_0	-1 (fixed)	-0.976 ± 0.021	-0.830 ± 0.058
w_a	0 (fixed)	0 (fixed)	-0.609 ± 0.228
$A_{\rm CMB}$	1.0008 ± 0.0026	1.0010 ± 0.0026	1.0006 ± 0.0024
$P_{\rm ACT}$	1.0026 ± 0.0025	1.0026 ± 0.0025	1.0023 ± 0.0025

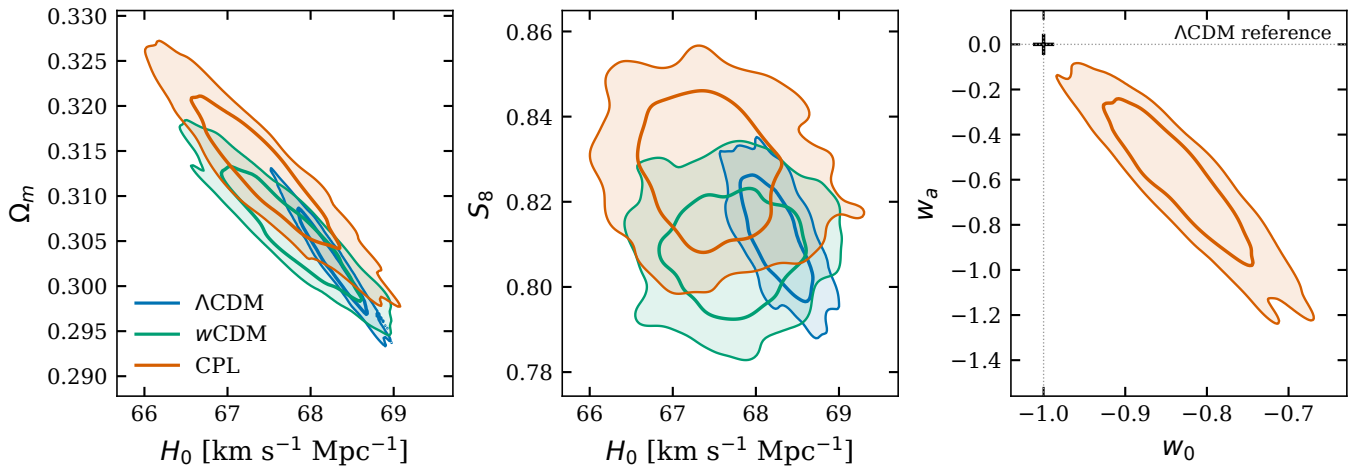


Figure 1. Direct-CAMB importance-corrected posterior constraints. Contours enclose 68 and 95 per cent of the plotted two-dimensional densities. In the CPL panel, the cross marks $(w_0, w_a) = (-1, 0)$. The displayed dark-energy covariance must be considered jointly; separate one-dimensional exclusions are not independent significance tests.

result, but a formal tension significance would require accounting for shared SN information and the full model dependence. The new GWTC-5.0 standard-siren analysis provides an independent distance-scale test with population and selection modelling (LIGO Scientific Collaboration et al. 2026); current broad siren constraints do not by themselves discriminate the two distance scales decisively.

Other compilations, including Union3, and cosmic chronometers merit future replacement tests with their intended likelihood conventions. Older overlapping BAO releases are not automatically additional information. Validated cosmological likelihoods from Euclid, Rubin and Roman would extend this analysis as they become available. A global analysis of all probes would additionally require their shared-volume and calibration covariances.

6 DISCUSSION AND CONCLUSIONS

This public-likelihood reanalysis combines current SN calibration and DESI Lyα information with Planck and ACT spectral data. The twelve independently seeded chains converge, and the retained direct-CAMB corrections preserve useful sample weight. The baseline ΛCDM result is $H_0 = 68.26 \pm 0.27\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. CPL improves

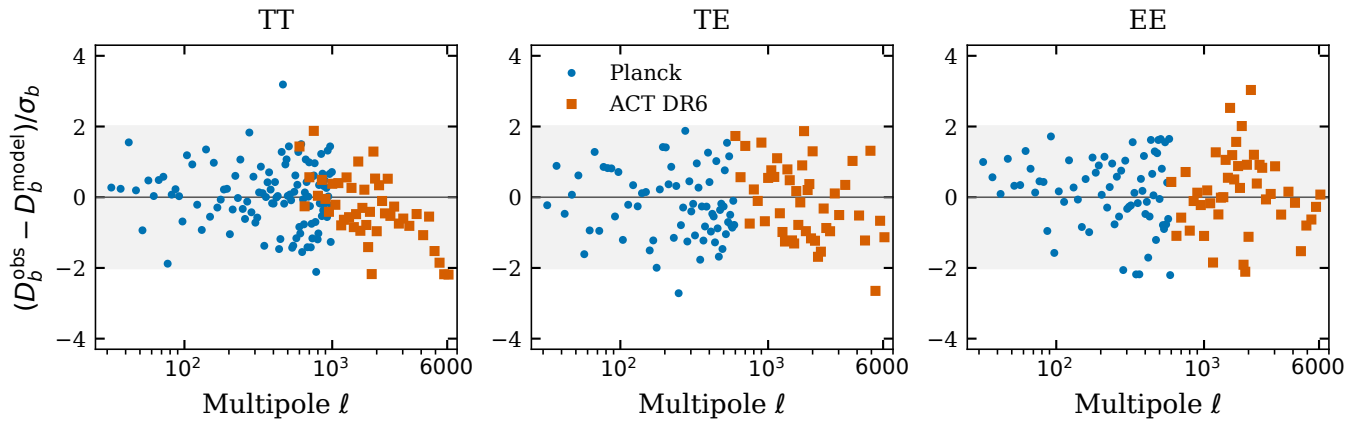


Figure 2. CMB bandpower residuals at the direct-CAMB local ΛCDM fit, divided by the released diagonal errors. Blue circles denote retained Planck bins and orange squares ACT DR6. The shaded region spans two diagonal standard errors. Bandpower correlations enter the likelihood; the displayed points are not independent unit-variance tests.

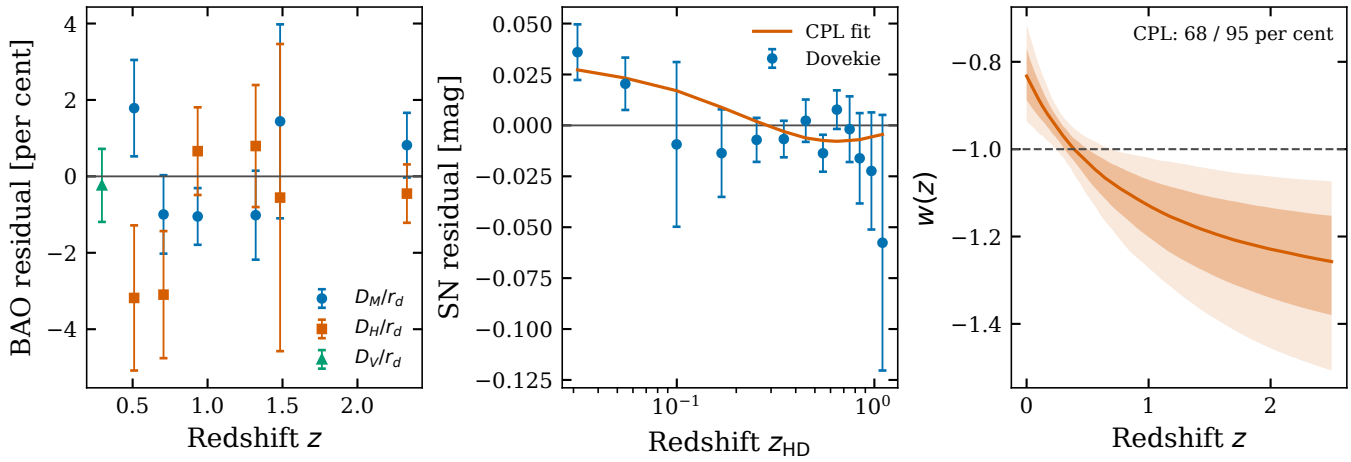


Figure 3. Distance and expansion-history tests. BAO residuals include the updated Lyα distance pair; the baseline ΛCDM prediction defines the reference. The supernova display uses generalized least-squares bins of Dovekie residuals, with covariance between bins retained in their construction; these points are for visualization, while the likelihood uses all 1820 objects. CPL $w(z)$ bands are posterior intervals within the stated parametrization.

Table 4. Direct-CAMB best evaluated local fit statistics. The calibration prior is included in D. Fixed likelihood offsets are retained; only same-data differences are interpreted. These are approximate local profiles, not certified global minima.

Contribution	ΛCDM	wCDM	CPL
Planck high-ℓ	220.98	221.16	220.79
ACT	160.17	161.63	157.01
Planck low-T	21.76	21.64	22.09
Planck low-E	396.05	396.07	396.02
BAO / AP	15.09	14.81	13.02
Supernovae	1634.77	1632.33	1629.46
Calibration prior	0.04	0.15	0.02
Total D	2448.85	2447.79	2438.41
ΔD	0.00	-1.06	-10.44
$\Delta D + 2\Delta k$	0.00	0.94	-6.44

the local deviance by 10.44, or 6.44 after the stated two-parameter penalty. This is an indication within a specified phenomenological model, not a Bayesian evidence calculation or a discovery claim. In this data combination, late-time dark-energy freedom moves H_0 farther from the Local Distance Network central value. A geometrical fit improvement therefore does not by itself resolve the distance-scale discrepancy. The essential limitation is model scope. Flatness, fixed neutrino mass and standard early-time physics are assumptions here. These chains cannot establish curvature limits, neutrino-mass bounds, early dark energy, modified gravity or the origin of any discrepancy. Late-time CPL freedom alone is not a general solution to a distance-scale mismatch, and w_0–w_a is an empirical description rather than a unique microphysical theory.

Before journal submission, an independent implementation should verify the likelihood conventions and compare alternative CMB reductions and full foreground treatments. Adding reconstructed lensing, SPT and galaxy-lensing likelihoods with validated cross-covariances is the natural next extension. The attached data, code, weighted draws and diagnostics make the present calculation inspectable and reproducible.

Table 5. Replacement and omission checks using direct-CAMB likelihood-ratio weights. Rows are shown only when weight ESS is at least 400 and Pareto $k < 0.7$. All changes preserve the baseline CMB treatment. These are alternative fits, not products of overlapping SN samples.

Alternative / model	$H_0\,[\mathrm{km\,s^{-1}\,Mpc^{-1}}]$	w_0	w_a	N_{IS}
Original Lyα BAO / ΛCDM	68.34 ± 0.28	-1 (fixed)	0 (fixed)	721
Original Lyα BAO / wCDM	67.79 ± 0.51	-0.976 ± 0.021	0 (fixed)	740
Original Lyα BAO / CPL	67.50 ± 0.58	-0.816 ± 0.058	-0.669 ± 0.231	574
Pantheon+ replacement / ΛCDM	68.30 ± 0.27	-1 (fixed)	0 (fixed)	785
Pantheon+ replacement / wCDM	67.87 ± 0.55	-0.982 ± 0.023	0 (fixed)	731
Pantheon+ replacement / CPL	67.68 ± 0.67	-0.864 ± 0.055	-0.496 ± 0.203	519
Supernovae omitted / ΛCDM	68.40 ± 0.29	-1 (fixed)	0 (fixed)	604

DATA AND CODE AVAILABILITY

The accompanying archive contains the pinned public likelihood inputs, their source URLs and SHA-256 hashes, the executed Python pipeline, random seeds, surrogate and direct-likelihood samples, numerical diagnostics, figure code and this manuscript's LaTeX source. Observational data and upstream likelihoods remain the work of their respective collaborations. No proprietary or mock observational data are used. The analysis was executed on 4–7 September 2026.

ACKNOWLEDGEMENTS AND DRAFT STATUS

This computational draft and its code were generated with ChatGPT at Nic Ross's request. The named recipient has not independently certified the analysis, and no authorship or institutional endorsement is asserted. We thank the survey collaborations and software authors for public data and likelihoods. This document has not been submitted to or accepted by MNRAS.

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APPENDIX A: CHAIN DIAGNOSTICS

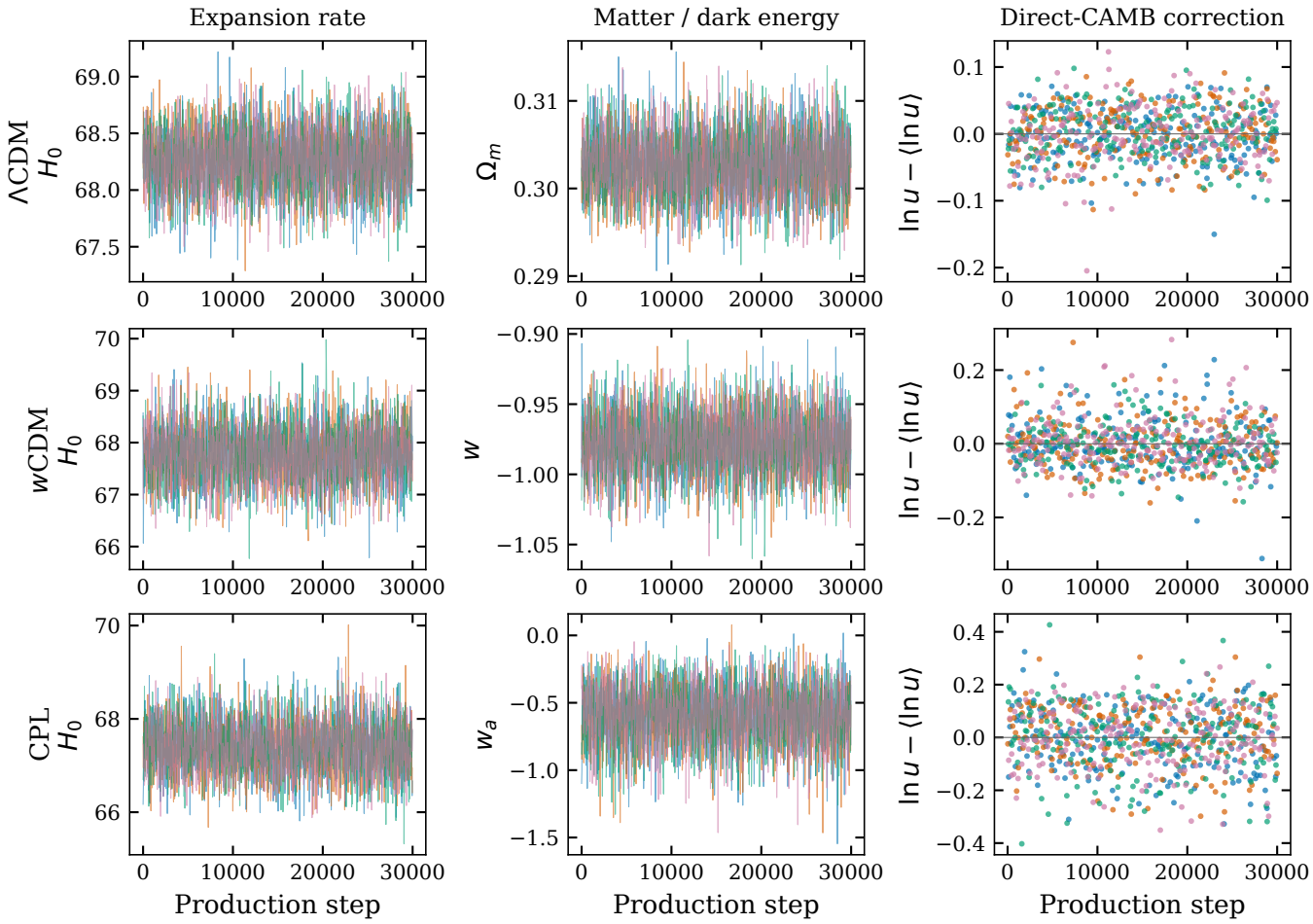


Figure A1. Production-chain traces and direct-likelihood importance corrections. Different colours denote independent chains. Display thinning is used only to keep the figure legible; diagnostic calculations use the full stored chains.